

Assignment 2c: Design Research Review

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Participants - Interviews

Interview Notes

P1 is a female diagnosed with Thoracic Outlet Syndrome, a condition where nerves and/or blood vessels are compressed near the shoulders, leading to symptoms like radiating pain, numbness and discoloration in the arm and fingers. Although she was officially diagnosed this past summer, she has experienced these symptoms for nearly two years, going through a long and frustrating process involving misdiagnoses across multiple states before finally receiving answers at UW. Her symptoms evolved over time, starting as back pain and progressing into more severe nerve-related issues, though manual therapy has helped her become mostly symptom-free on many days. She identifies overhead movements and certain physical activities as triggers, which has impacted her ability to continue her hobbies like tennis, pickleball, and painting. She is currently working in a lab and has expressed issues with that. While she has not used health tracking apps; she expressed interest in better understanding patterns such as time of day or weekly trends that may influence her symptoms. She also pointed out that current tracking methods, like the pain scale, feel inadequate since they lack personal context. Overall she is open to health data tracking but emphasized how she feels the importance of ethical use and transparency in personal data should be prioritized.

P2 is a 19 year old female with chronic abdominal pain for at least 3 years. If she could know anything about her condition, she would want to know what she could and couldn't eat, and why. During flare-ups, she clears her schedule and uses short-term treatments (lights off, ice packs, ginger, Tylenol, etc.). The main challenges with this are being behind on homework, missing out on time with friends, and picking what to eat. She usually pushes through normal symptoms, although it makes normal functions more difficult and just makes the next crash worse. Symptoms have changed over the years. The process of finding "safe foods" (and avoiding triggers) is one of trial and error, and stick with it, all while trying medication and lifestyle changes. She wishes that she could predict the crashes, so she could better prepare for them. She was the only interviewee that said that she couldn't describe her symptoms to doctors accurately. She would like to see her health data, has tried other apps, but doesn't want to pay to see her own data. They helped with tracking meds, but were too cumbersome and asked for too much information daily. Also, they were too clinical and bland in aesthetic. She does not track her health data currently. She'd like to know what foods she can and can't eat. She wants platforms to be respectful of her data and not limit it from herself.

P3 is a 21 year old female with chronic face and head pain, sometimes radiating to head and back. This is a minor flaw in our interviews; all interviewees are college-age females. She has had this condition for about 7 years. Our interviewees had a good range of length of condition. If she could know anything about her condition, she would want to know what the risk factors are and what caused it. This is a through line with all interviewees – they would like to know what caused their conditions. Bad days / flare-ups look like laying in bed, unable to do other activities.

These flare ups are navigated based on energy level and how bad the flare up is. Flare ups are very distracting, annoying to deal with. Over the years, the pain has gotten worse, and spread to new places. This is something all the interviewees had in common; all their conditions have changed over time. She has tried every class of medication, as well as all the lifestyle changes. Nothing has helped, and diets made it worse. She would like to know more about triggers, but has already tried many pain diaries. She is now able to describe her symptoms pretty accurately, but had to learn how to describe it over time. Allodynia affects her the most day-to-day, having to pick outfits hairstyles, etc. that minimize pain. She'd be interested in sleep vs. pain intensity vs. mood vs. function. She has tried multiple health tracking apps, which did not help her. She noted that the main failure of the tracking apps is that they were both boring and took too long daily. Filling something out daily for 5+ minutes can be daunting, especially during a flare-up. She would like to know pain intensity, function and sleep quality data. Pain diaries usually have a human outline to mark pain, and she would like it if she was able to mark more specific spots, as well as the specific types of pain. She feels comfortable with a HIPPA-compliant app, storing and analyzing data with permission. 3

Participants - Social Media Survey

As discussed with course staff during section, for our second design research method, we posted on the social media platform Reddit asking/surveying people with chronic conditions about their experiences. We posted in various subreddits like r/AskReddit and r/ChronicIllness. Reddit keeps users anonymous, and due to various subreddit guidelines, we were unable to gather personal background details other than the information users volunteered in their responses. However, most Reddit users in the subreddits we posted in are helpful (except a few who responded to our posts with troll or mean comments) and in subreddits like r/ChronicIllness, we observe that most people posting/commenting are ones with chronic conditions who possess specialized knowledge in their own lived experiences. Additionally, these users shared that they have various degrees of experience with tracking their health data.

Themes - Interviews

People living with chronic conditions are caught in a frustrating cycle of trial and error. Participants spent years seeking diagnoses and identifying triggers, such as specific movements, foods, or sleep patterns. They are deeply interested in seeing how these lifestyle factors correlate with their symptoms, yet they find current tracking tools inadequate. Many existing apps are described as "soulless," clinical, or too time-consuming to use during a flare-up. When pain is severe, any task taking longer than 5 minutes becomes a burden. This suggests that design must prioritize low-friction, rapid logging.

A major point of friction is the disconnect between medical metrics and lived experience. Users find standard 1–10 pain scales annoying because they lack personal context, like whether a user can do their hair, finish homework, or participate in hobbies. They want tools that allow for specific anatomical marking and the ability to track "functionality" and mood alongside physical pain. They also want predictive features that could help them anticipate a bad day so they can adjust their schedules in advance.

Finally, while many participants are open to sharing their data to improve their care or contribute to research, privacy is still a significant concern. Users are wary of their health information being sold or used against them in professional or insurance contexts. They want transparent, ethical platforms that give them full control over who sees their data. These findings suggest that an ideal design should move away from rigid, clinical forms and toward a flexible, community-oriented tool that focuses on identifying personal triggers, predicting energy crashes, and improving the quality of communication between patients and their doctors.

Themes - Social Media Survey

Through searching and posting health tracking app related topics on Reddit, there are themes across three dimensions: types of data, current challenges and functionality users favor. For types of data, it appears that sleep data tracking is crucial because some users said apps could help them to identify their sleep patterns with which some symptoms could be explained by sleep patterns. For functionality users favor, building community with people with similar conditions, linking and synthesizing medical records and other medical information, and simplicity & customizability UI (eg. Google Sheet) are shared characteristics of what people like about existing solutions. For the current challenges, some users have privacy concerns over their health data, especially how those data could be used to go against users in certain situations. In addition, some people found manual symptom tracking cumbersome.

Task Analysis Questions

1. Who is going to use the design?

The users of this design are individuals with chronic illnesses. It will be an app dedicated to those who wish to identify lifestyle changes, triggers, and trends based on their daily health data to improve their quality of life

2. What tasks do they now perform?

The tasks users perform include logging in to the platform with their login credentials and entering their daily health symptoms.

3. What tasks are desired?

The tasks users want are the ability to see different trends in their health when they notice certain symptoms and to better understand what future steps they can take.

4. How are the tasks learned?

We want to make the platform very user-friendly so that individuals with different technology contexts can learn to navigate it. Having many resource pages on this platform would be ideal, so individuals don't get lost navigating it.

5. Where are the tasks performed?

The tasks are performed on the platform itself. This will be securely available to each individual's account, ensuring privacy is a top priority.

6. What is the relationship between the person and their data?

Personal health data, access restricted to only the user, maybe their provider (HIPAA)

7. What other tools does the person have?

Other health apps, journals

8. How do people communicate with each other?

This will be very personalized to each individual. We want to make sure that all data entered into the app is accessible only to the individual and that there is no ability to transfer it without security measures. One thing that can be used is for patients to show their symptoms and trends to doctors to accurately describe what they are going through.

9. How often are the tasks performed?

The tasks can be performed hourly, daily, or weekly, depending on what the user is specifically looking for. We want to give users the ability to track on their own timeline and under their own conditions, making it less chore-like.

10. What are the time constraints on the tasks?

Routine entry of health symptoms should take a short time (<5 minutes as mentioned during interviews) to minimize the daily burden.

11. What happens when things go wrong?

Lifestyle changes, triggers, and trends are either not identified or incorrectly identified.